

CS364 Semester Project

Project Proposal

Abstract

For this project, I want to create a Football Performance Database. This database will be designed to store and analyze detailed sports data from the NFL. The goal is to provide insights into team and player performance. This can help fans, analysts, and coaches evaluate which positions are excelling and which teams have the most efficient offense and defense. The database will store key statistics for players and teams, including performance metrics such as yards, touchdowns, and player efficiency ratings. It will also be capable of calculating advanced statistics like completion percentages, rushing efficiency, and turnover ratios that offer deeper football insights.

The system will allow users to run complex queries to assess the best-performing positions, offensive and defensive units, and much more. Users will also be able to compare performance across different time frames, such as single games or full seasons, to identify trends and patterns. The database will feature statistical breakdowns by player and team, enabling a more comprehensive analysis of game outcomes. This will hopefully also be insightful when trying to predict outcomes of future games based off past trends and statistics.

The Football Performance Database aims to be a valuable tool for football analysis. Users will be able to generate reports and visualizations that highlight performance trends over the course of a season or within individual games. This will hopefully allow its audience to examine football and sports from a more analytical, data-driven perspective, which could influence the evolution of strategies and decision-making within the game itself.

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For this project, I aimed to create a Football Performance Database to store and analyze detailed NFL data. Initially, the goal was to include comprehensive statistics for every NFL team, player, and coach, allowing for evaluation of team and player performance, such as identifying the best-performing positions and the most efficient offenses and defenses. Additionally, the database was intended to enable users to query performance trends across various time frames and generate reports and visualizations for analytical purposes.

However, as the project progressed, I realized the massive scope of including statistics for every team, player, and game across the NFL. To create a manageable and functional prototype, I narrowed the database to include complete data for teams in the NFC North. This includes detailed records for every team, player, and coach in that division.

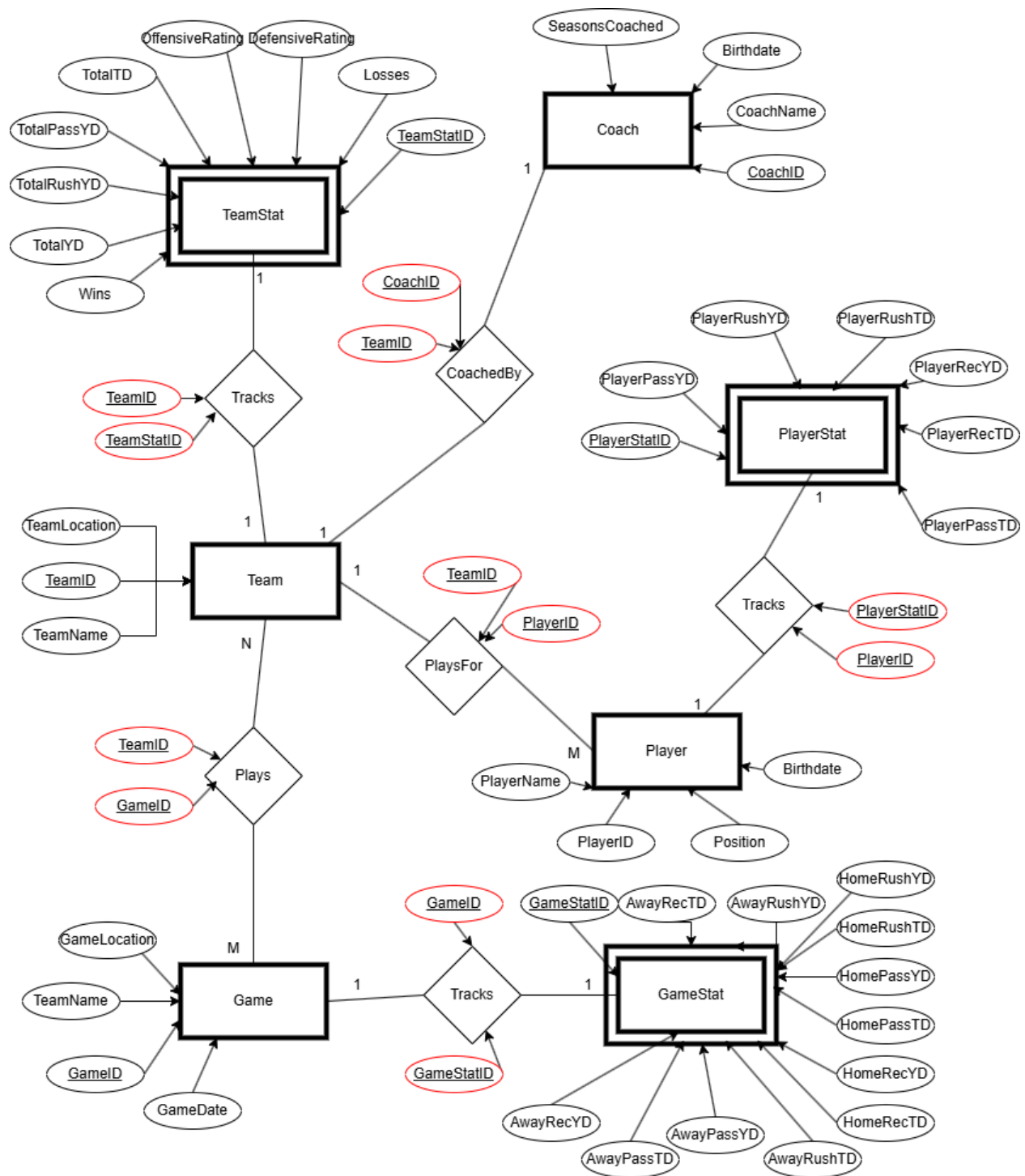
I also developed a graphical user interface (GUI) using WindowBuilder in Eclipse. The GUI allows users to:

- View all coaches, players, or games for a specific team using combo boxes, with results displayed in a table.
- Perform updates, insertions, and deletions of players.
- Query specific players and display their statistics in a secondary results table.

Although the final product does not encompass the entire NFL as originally planned, the database is thorough for the NFC North and provides a good foundation. The combination of the database and the GUI demonstrates the potential for comprehensive football data analysis, providing valuable insights into player and team performance within a manageable scope. Overall, the project showcases the possibilities of applying data-driven analysis to sports.

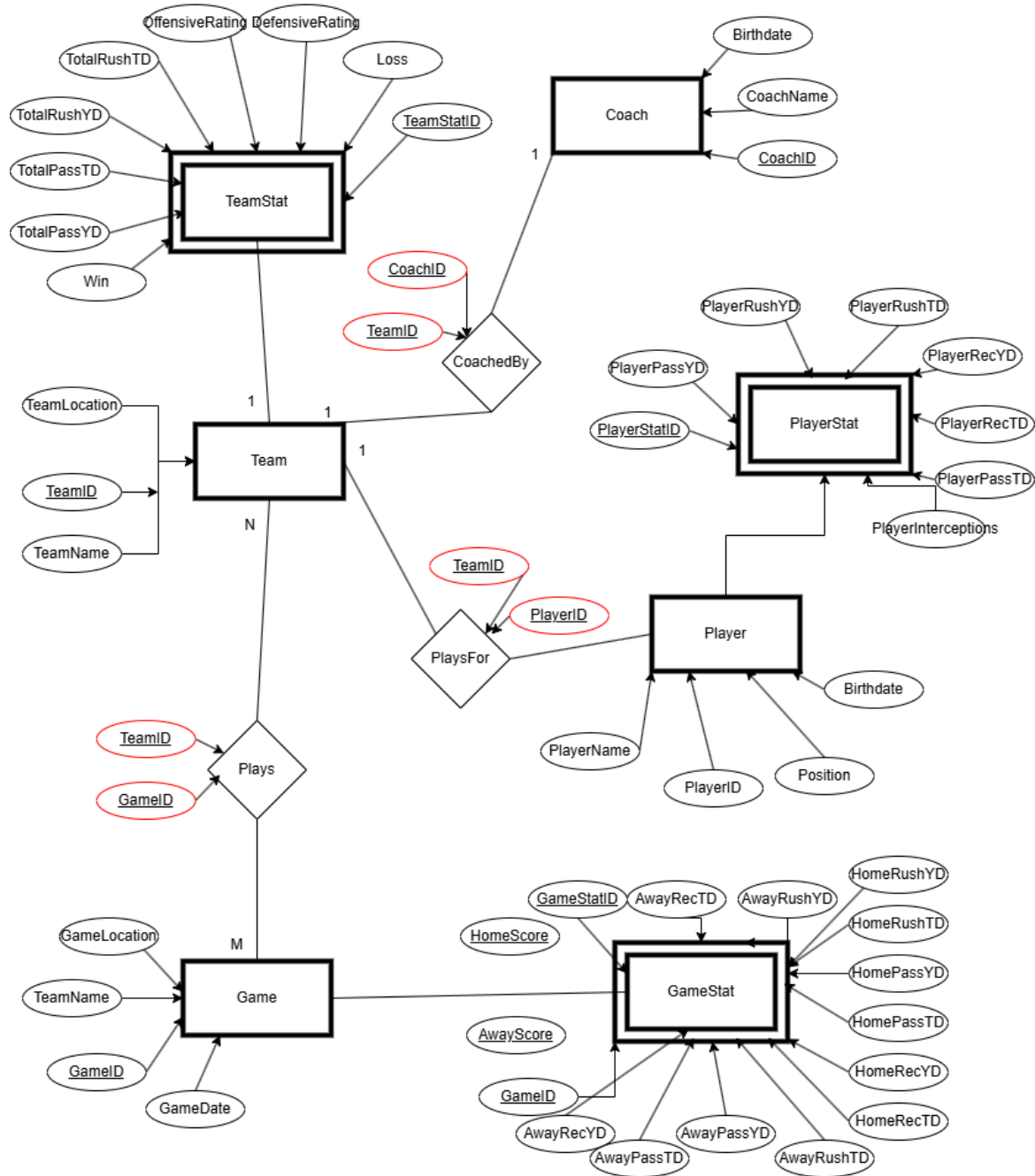
ER Diagram

We can create an ER diagram to help visualize what the Football Performance Database will look like from a conceptual schema standpoint. To represent the database, we will use 4 entities coach, team, player and game and 3 weak entities to track stats for team, player and game. Our main entities hold generic information about the entity while the statistic entities hold various football stats which we have interest in and would like to run queries on. We have various relationships as represented by diamonds in our diagram. The team entity has a relationship with virtually every part of our database and as you can see all the main entities have a relationship with their stat entities. We have IDs for all entities which will allow us to ensure easy access to specific players, coaches, teams or games. Foreign keys for relationships are represented in the ER diagram by red ovals and they are linked to the relationship.



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NFL Database ER Diagram



From the beginning to the end of the project, the schema of the project would change to create a more functional program or to help make the database easier to navigate. The ER diagram demonstrates the features necessary for analyses which ensure the database is functional and supports meaningful queries. Some relationships with the database which deemed as unnecessary and removed from the scheme. This includes some of the relationships between weak entities.

Functionality

1. Player and Team performance tracking
 - a. Store and update statistics for players and teams after each game
2. Game data tracking
 - a. Store and update game specific statistics including scores, dates and teams
3. Player and team comparisons
 - a. Allows user to compare individual's player performance and team performance
4. Report Generation
 - a. Create reports that summarize player and team performance like trends over recent games

Some queries might include

- Top 5 players by position
- Best offensive teams
- Most yards by a player in a single game
- Player with the most touchdowns who is younger than the league average
- Team with the best record when playing at home

The possibilities for queries are endless as the user can ask any questions that align with the statistics being tracked in the database.

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The database allows users to achieve almost all the intended functionality, but it is limited by the GUI and my window builder knowledge. Some queries that you can run on the database include.

Queries For Project

Group 1 aggregate functions, LIKE, GROUP BY, ORDER BY, LIMIT

This Query selects the top 5 quarterbacks in the NFC North and orders them by total touchdowns

- ```
SELECT P.PlayerName, P.Position, SUM(PS.PlayerRushTD + PS.PlayerPassTD + PS.PlayerRecTD) AS TotalTD
FROM Player as P
JOIN PlayerStat as PS ON P.PlayerID = PS.PlayerID
WHERE P.Position = 'QB'
GROUP BY P.PlayerID
ORDER BY TotalTD DESC
LIMIT 5;
```

|   | PlayerName     | Position | TotalTD |
|---|----------------|----------|---------|
| ▶ | Jared Goff     | QB       | 17      |
|   | Sam Darnold    | QB       | 17      |
|   | Jordan Love    | QB       | 15      |
|   | Caleb Williams | QB       | 9       |
|   | Malik Willis   | QB       | 4       |

```

Group 2 HAVING, OFFSET, outer join, joining four or more tables
This query selects 15 players in the nfl who have won the least amount of games this season
because this database only takes into consideration the NFC North the output is the all of the
players on the chicago bears

```

```

SELECT T.TeamName, P.PlayerName, COUNT(PL.GameID) AS GamesWon
FROM Team T
JOIN Plays PL ON T.TeamID = PL.TeamID
JOIN PlaysFor PF ON T.TeamID = PF.TeamID
JOIN Player P ON PF.PlayerID = P.PlayerID
WHERE PL.Result = 'W'
GROUP BY T.TeamID, P.PlayerID
HAVING COUNT(PL.GameID) > 3
ORDER BY GamesWon ASC
LIMIT 15
OFFSET 0;

```

|   | TeamName          | PlayerName       | GamesWon |
|---|-------------------|------------------|----------|
| ► | Chicago Bears     | Caleb Willaims   | 5        |
|   | Chicago Bears     | DAndre Swift     | 5        |
|   | Chicago Bears     | Roschon Johnson  | 5        |
|   | Chicago Bears     | Khalil Herbert   | 5        |
|   | Chicago Bears     | DJ Moore         | 5        |
|   | Chicago Bears     | Rome Odunze      | 5        |
|   | Chicago Bears     | Keenan Allen     | 5        |
|   | Chicago Bears     | Cole Kmet        | 5        |
|   | Chicago Bears     | Gerald Everett   | 5        |
|   | Green Bay Packers | Josh Jacobs      | 6        |
|   | Green Bay Packers | Emanuel Wilson   | 6        |
|   | Green Bay Packers | Jayden Reed      | 6        |
|   | Green Bay Packers | Christian Watson | 6        |
|   | Green Bay Packers | Romeo Doubs      | 6        |
|   | Green Bay Packers | Dontayvion Wicks | 6        |

```

Group 3 subqueries, IN, set operators
This query obtains all players in the NFC North who have rushed for over 500 yards and have over 5 touchdowns and orders them
in descending order by touchdowns
SELECT P.PlayerName,PS.PlayerRushYD,PS.PlayerRushTD
FROM Player P
JOIN PlayerStat PS ON P.PlayerID = PS.PlayerID
WHERE P.PlayerID IN (
 SELECT PF.PlayerID
 FROM PlaysFor PF
 JOIN PlayerStat PS2 ON PF.PlayerID = PS2.PlayerID
 WHERE PS2.PlayerRushYD > 500
 AND PS2.PlayerRushTD >= 5
)
ORDER BY PS.PlayerRushTD desc;

```

|   | PlayerName       | PlayerRushYD | PlayerRushTD |
|---|------------------|--------------|--------------|
| ▶ | David Montgomery | 520          | 8            |
|   | Jahmyr Gibbs     | 727          | 7            |

## Stakeholders

There are many different types of users that may use the football performance database. This includes coaches, other analysts and fans. Coaches can use the database to analyze both team and individual performance. They can run queries to assess which positions or players are excelling, and which areas need improvement. Coaches may focus on offensive and defensive rating, player efficiency stats and game performance to make strategic decisions. Analysts can use the advanced statistics stored in the database to try and evaluate trends and predict future performance. Like coaches they can use queries to compare players and teams based on historical data. Fans can use the database to track the performance of their favorite players and teams. Fans will primarily use the report and visualization to get quick insights into game stats. Another group that could use the database includes journalists who are reporting and writing about games. They can use the database to compare players and strengthen their takes by using the statistics found in the database.

After Constructing of the database and the gui the target audience would still be similar. The database might not be as popular of course because of its limited functionality but I think that it would appeal to specifically NFC North football fans. This is convenient since they are the best division in the NFL and by far the most exciting.

## Technological Requirements

This application will be implemented as a web-based platform to make it accessible to a wider audience, including fans, analysts, coaches and other stakeholders. For programming languages on the backend, we will be using Java for logic and data processing. For the front-end user interface, we will be using Python to create our website and gui. As a database for our platform, we will be using MySQL as the database management system, and we will be connecting

to it via JDBC. This will allow us to run SQL queries from the backend to retrieve data. For sharing code, we plan to use email directly with all our code in a singular zip file. For this project I will be working alone because this is something I've wanted to do for a while, and I want to really understand it. My experience includes a good foundation for java and SQL. This will be my first time building a website using python which I have beginner level experience in. I plan to use Django to build my website using python.

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After writing my original project proposal and while creating the initial foundation of my database I thought about how I wanted to implement the gui portion of the project. I ended up deciding on using the window builder because of my familiarity with coding and java and my unfamiliarity with building websites. This project uses SQL to store and query a database that represents players, statistics, teams and games within the NFL. The database is connected to a java program via JDBC which creates a Window Builder which allows user to query the database in the allowed fashion of the gui. This includes being able to look up coaches, games or players by team, inserting, updating and deleting players, and being able to look at specific players stats.

## Screenshots

After the project demo I added a section of the gui that allows the user to put in a variety of player stats, select how they would want the data to be ordered, and run a query on the database showing 10 players who have more of all particular stats ordered by the users preference.

The screenshot displays a Java Swing window titled "NFL Database" with a light gray background. It is divided into several functional sections:

- Manage Database:** Located at the top right, it contains a subtitle "Fill in the boxes below with content of the desired format and select an action". It has input fields for "Player Name:", "Birthdate (YYYY-MM-DD):", "Position:", and "Player ID:". Below these are "Insert", "Update", and "Delete" buttons.
- Player Stats:** Located in the middle left, it features a "Select Player" dropdown menu with "DJ Moore" selected. Below the dropdown is a table showing stats for DJ Moore.
- Advanced Player Queries:** Located at the bottom, it includes an "Order By" dropdown menu set to "PlayerRush...". Below this is a table with columns for various stats (RushYD, RushTD, PassYD, PassTD, RecYD, RecTD) and a "Run Q..." button.

**Player Stats Table:**

| Name     | ID | RushYD | RushTD | PassYD | PassTD | RecYD | RecTD | Interceptions |
|----------|----|--------|--------|--------|--------|-------|-------|---------------|
| DJ Moore | 47 | 18     | 0      | 0      | 0      | 398   | 3     | 0             |

**Advanced Player Queries Table:**

|                  | RushYD | RushTD | PassYD | PassTD | RecYD | RecTD |
|------------------|--------|--------|--------|--------|-------|-------|
| Josh Jacobs      | 762    | 3      | 0      | 0      | 128   | 1     |
| Jahmyr Gibbs     | 727    | 7      | 0      | 0      | 202   | 1     |
| Aaron Jones      | 653    | 2      | 0      | 0      | 258   | 1     |
| DAndre Swift     | 564    | 4      | 0      | 0      | 230   | 0     |
| David Montgomery | 520    | 8      | 3      | 1      | 185   | 0     |
| Jayden Reed      | 110    | 1      | 0      | 0      | 620   | 3     |



### NFL Database

Select Team

Green ...

Select Enti...

Player

Run Q...

|                |    |            |    |
|----------------|----|------------|----|
| Jordan Love    | 12 | 1998-11-02 | QB |
| Malik Willis   | 12 | 1999-05-25 | QB |
| Josh Jacobs    | 12 | 1998-02-11 | RB |
| Emanuel W...   | 12 | 1999-05-08 | RB |
| Jayden Reed    | 12 | 2000-04-28 | WR |
| Christian W... | 12 | 1999-05-12 | WR |
| Romeo Do...    | 12 | 2000-04-13 | WR |
| Dontayvion ... | 12 | 2001-06-16 | WR |

Reset

### Manage Database

Fill in the boxes below with content of the desired format and select an action

Player Name:

Elliot Maringer

Birthdate (YYYY-MM-DD):

2003-12-12

Position:

WR

Player ID:

999

Insert

Update

Delete

### Player Stats

Select Player

Caleb Willai...

| Name           | ID | RushYD | RushTD | PassYD | PassTD | RecYD | RecTD | Interceptions |
|----------------|----|--------|--------|--------|--------|-------|-------|---------------|
| Caleb Willaims | 43 | 236    | 0      | 1785   | 9      | 0     | 0     | 0             |

Run Q...

### NFL Database

Select Team

Chicago... ▾

Select Enti...

Game ▾

Run Q...

|   |                     |   |
|---|---------------------|---|
| 1 | Chicago, Illinois   | W |
| 2 | Houston, Texas      | W |
| 3 | Indianapolis, India | L |
| 4 | Chicago, Illinois   | W |
| 5 | Chicago, Illinois   | W |
| 6 | Chicago, Illinois   | W |
| 8 | Landover, Maryla    | L |
| 9 | Glendale, Arizona   | L |

Reset

### Manage Database

Fill in the boxes below with content of the desired format and select an action

Player Name:

Josh Allen

Birthdate (YYYY-MM-DD):

Position:

Player ID:

Insert

Update

Delete

### Player Stats

Select Player

David Mont... ▾

| Name             | ID | RushYD | RushTD | PassYD | PassTD | RecYD | RecTD | Interceptions |
|------------------|----|--------|--------|--------|--------|-------|-------|---------------|
| David Montgomery | 90 | 520    | 8      | 3      | 1      | 185   | 0     | 0             |

Run Q...

NFL Database

Select Team

Minnes... ▼

Select Enti...

Coach ▼

Run Q...

Kevin OConnell

1985-05-25

Reset

Player Stats

Select Player

Jayden Reed ▼

Manage Database

Fill in the boxes below with content of the desired format and select an action

Player Name:

Jordan Love

Birthdate (YYYY-MM-DD):

2024-12-10

Position:

WR

Player ID:

Insert

Update

Delete

| Name        | ID  | RushYD | RushTD | PassYD | PassTD | RecYD | RecTD | Interceptions |
|-------------|-----|--------|--------|--------|--------|-------|-------|---------------|
| Jayden Reed | 102 | 110    | 1      | 0      | 0      | 620   | 3     | 0             |

Run Q...